

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EEE 324 — Past Questions & Materials

by Renance

Course: EEE 324

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 2

Levels: 300

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Paper 1 of 2 — EEE 324 - 2023/2024 - Second Semester - Applied Electronics I

Level 300 • Second Semester • PastQuestion

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COLLEGE OF TECHNOLOGY & ENGINEERING

Second Semester Examination 2023/2024 Session

Course Title: Applied Electronics | (EEE 324)

Time Allotted: 2 Hours Course Unit: 2

Instruction: Answer any four questions. All assumptions should be clearly stated and please do not write at

the back or front of the question paper. Do your rough works at the back pages of your booklets

QUESTION ONE

(i) write the expressions for A, B, C and,

(ii) Determine the expression for V_o/V_i . (17.5 marks)

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(iit) Calculate the current, I_1 , and I_2 , of the equivalent circuit draw (17.5 marks)

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Figure 3: Radio Frequency Oscillator

QUESTION FOUR

Figure 4 is the Widlar current source also tagged as differential amplifier circuit here. Given that $I_{C1} = 20\text{mA}$, $\beta = 6$,

$\beta = 90$, $\alpha = 0.9$, $V_z = 24.5\text{ mV}$, $V_{ig} = 0.4\text{ V}$. Calculate the:

(i) Reverse saturation current (I_{cs}).

(ii) What is the value of the differential voltage given that $R_e = 0.6\text{ k}\Omega$

(ii) Calculate the value of I_{p2} .

(17.5 marks)

Figure 4: Widlar current source (differential amplifier)

T_p -loss Cr 3) V_e

QUESTION FIVE re 2

Given that $V_c = 4\text{V}$, $I_{nss} = 12\text{ mA}$ of a given Field effect transistor (FET), deploy the Shockley's equation to:

a (i) Determine the value of V_p when $I_p = (0.6 I_{nss})$, V_{as}

(33) Calculate I_p and V_p of a FET at $I_p=0.8$, $I_p= (0.2 I_{pss})$, $I_b= (0.5 I_{pss})$, $[b= (0.16 I_{pss})]$, $I_b= I_{pss}$, by applying

Shockley's equations, and plot the graph of I_p against V_p . (17.5 marks)

QUESTION SIX

Figure 5 is the equivalent circuit diagram of a combined low pass and high pass filter. Given that $V_i = 20$ V, $R_i = R_2$

$= 25\Omega$, $f = 50$ Hz, $C_1 = 20$ pF, $C_2 = 15$ pF, $Q = 0.7$. Calculate the:

i (i) Voltage (V_o)

(ii) Output voltage (V_o)

iii) Ratio of V_o to V_i , and ;

. V_j to V_s . (17.5 marks)

A; Afsts

ii T Figure 5: Equivalent circuit diagram of

7 Raf] 1/8 all Ye combined low and high pass filter

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f) A cargo ship with an area of 20000: is to be protectec!. If the current density of the ship steel material

is 0.15mA/m', with the coating breakdown factor of 0.06. Magnesium anodes with dimensions 0.5m by

0.15m are to be employed. Assuming the anode resis'ivity to the environment is 30 ohms.cm, anode

electrochemical potential is -1.7 and that of the ship hull is -0.75, Determine the number of anode required for the ship protection. (5 Marks)

QUESTION 6 (15 Marks)

8) Describe the four main categories of corrosion control methods. For the jacket platform's immersed

zone, explain which two of these methods are used in combination and why. (2 Marks)

o) A corrosion coupon retrieved from a produced water pipeline on the platform shows a higher-than-

'expected corrosion rate. Describe what a corrosion coupon is and how it functions. What will be the

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Paper 2 of 2 — EEE 324 APPLIED ELECTRONICS I

Level 300 • Second Semester • PastQuestion

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 DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
 Second Semester Examination 2024/2024 Session Bes pe
 Course Title: Applied Electronics (EEE 324)
 Time Allotted: 2 Hours Course 1

Instruction: Answer any four questions. All assumptions should be clearly stated and please do not write at the back or front of the question paper. You can do your rough works at the back of your sheet.

1. Draw the circuit symbol or schema of the following Bipolar Junction Transistors (BJTs)

(i) Common Base (2 marks),

(ii) Common Emitter (2 marks),

(iii) Common Collector (2 marks).

(iv) Derive the analytical (Mathematical) relationship among the common base-common emitter-common collector gains (α , β) of the BJTs (8 marks).

2. (a) Draw a common source small signal model of a Field Effect Transistor (FET) low frequency model (2 marks)

(b) Derive the drain current (I_D) of the FET model above (5 marks).

(c) Convert the small signal FET low frequency architecture in (a) to the Voltage Controlled Voltage source model (2 marks).

(d) Derive the Voltage gain A_v equation of the FET model in question (c) above by using Thevenin's theorem (5 marks).

(e)

(f)

3. Compute the: (i) input voltage gain (A_{vi}) (5 marks), (ii) output voltage gain (A_{vo}) of the (CS-CS) FET shown in Figure 1 (5 marks). Find the output voltage across a load R_L connected across the two-stage FET circuit in Figure 1 given an input of 20 mV (4 marks). Assume identical transistors with $g_m = 2.5 \text{ mS}$ and $r_{gs} = \infty$.

$V_{DD} = 20 \text{ V}$

$R_{in} = 3 \text{ k}\Omega$

$V_{in} = 20 \text{ mV}$

(b) Find the output voltage across the load R_L (5 marks).

Figure 1; A two-stage FET circuit

4. Compare the linear ideal amplifier to the a well-labelled practical model of:

(i) Impedance amplifier (3 marks),

Gi) Transconductance amplifier (3 marks).

® (ii) Voltage amplifiers (3 marks). F : : :

Gy) By applying the Thevenin's or-Nortan's theorem. derive the mathematical equation of the gains of (i) to (ili) (5 marks)

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Course Title: Applied Electronics | (EEE 324) tee

Time Allotted: 2 Hours Course Unit: 2

5, Figure 2 represents a feedback amplifier of a FET transistor,

(i) write the expressions for A, B, C and

(ii) Determine the expression for V_o/V_i ,

+ A

' < a Me

c B

0.01

Figure 2: Feedback Circuit diagram

6, Figure 3 is the schematic circuit symbol of a JFET voltage divider.

(0) Draw the equivalent circuit of the voltage divider (2marks),

Gi) Compute the equivalent Voltage (V_{go}) and Resistor (R_g) (6 marks),

(ij) Compute the value of I_p @ marks).

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7. Given that $V_p = -5V$, $I_{pss} = 12\text{ mA}$ of a FET,

@) Use the Shockley's equation to the determine the value of V_{os} when $I_p = (I_{nss}/8)$ (2 marks)

Gi) Determine I_p and V_{es} of a FET at $I_p=0$. $I_n = (I_{pss}/16)$, $I = (I_{pss}/8)$, $I_p = (I_{nss}/4)$, $t = \epsilon$ $I_{ngs}/2$, $I_p = I_{ps}$, by applying Shockley's equations (7 marks).

(ii) Plot the graph of I_p against V_{os} from (ii) (6 marks).

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